

THE COMPLETE DECK

Every Repo

Two hundred and twenty-six projects across twenty-six layers — the full assembly list, not the short one. Each entry says what it is, what job it does in *your* build, what it is licensed under, and whether to use it, read it, handle it carefully, or leave it alone.

- CORE

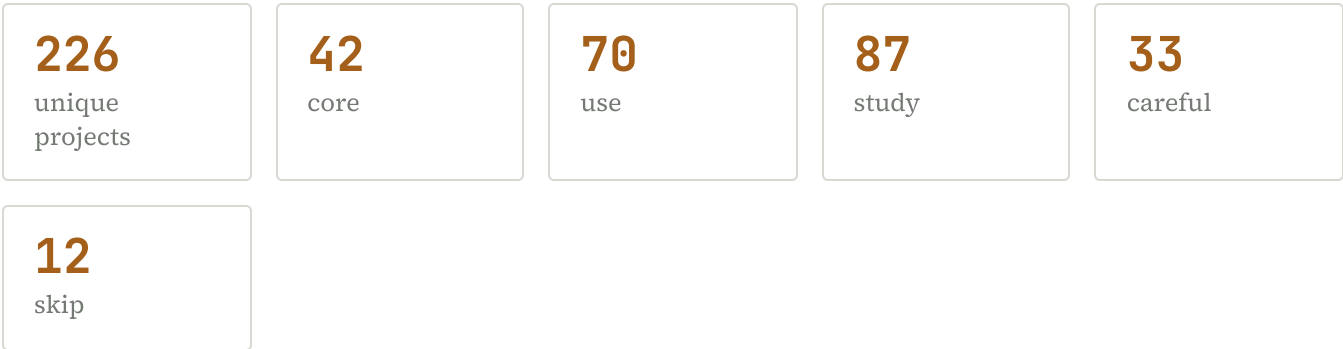
Load-bearing
- USE

Build on this
- STUDY

Read it, don't depend on it
- CAREFUL

Real catch attached
- SKIP

Wrong tool, or a trap



Contents

Twenty-six layers

Ordered roughly from the metal upward: the box, the distribution machine, the execution layer that is the actual product, then the platform above it.

00	Your own repositories	14	01	The OS image and the update spine	13
02	The comparable, read closely	4	03	Boot chain and hardware root of trust	7
04	Apps on the box, and the naming trap	5	05	The kiosk surface	4

06	Containers, sandboxes and the isolation ladder	16	07	Registry, fleet and entitlement — the Red Hat machine	11
08	The pipeline and the policy gate	5	09	Supply chain: signing, transparency, SBOM, scanning	11
10	The execution layer — remote control of a real phone	24	11	The other two execution paths	7
12	Their screen, on their phone	3	13	Agent protocols and the orchestration layer	13
14	Workflow, events and durable execution	12	15	Models and inference — the terabyte of specialists	20
16	Voice	13	17	The data plane	9
18	Getting the data in — email as the universal API	5	19	Identity, authorization, policy, secrets	6
20	Reaching the box without owning it	3	21	Money	4
22	The app store layer	11	23	Surfaces — what the customer and the world actually see	4
24	Read, do not adopt — the IBM and Red Hat shelf	4	25	The live traps	16

How to read the licence column

Licences are recorded as they stood when this project catalogued them, and they change — three entries here changed after they were first written down. A ✓ beside a repository name means that entry was re-verified on the web today; everything else is from the working record and should be re-checked before it becomes load-bearing in a build you ship. GitHub itself is unreachable from the environment this was compiled in, so nothing here was confirmed by opening the repository directly.

Your own repositories

The fourteen that already exist under CultureReset. These are the things you are not cloning from anyone, and the reason the rest of this list is an assembly job rather than a build-from-zero. Names as they appear in the session; rename freely.

Verdicts here are about role in the assembly, not code quality — I have not read these repositories, only the session's description of them.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
CultureReset/the-ghost-ai-	—	CORE	The box. Where this record, the platform assets files and the flow/runner/pack code live. The cut from.
CultureReset/cybercheck-core	—	CORE	The private truth. Business record, observation provenance. The one place data is authoritative never published from.
CultureReset/cybercheck-node	—	CORE	The node runtime. Runs identically on the box. Same code, different address — this is what runs the product and not a demo.
CultureReset/cybercheck-cloud	—	CORE	The hosted half. The node when the customer's entire first twenty-five customers live here because of the cloud.
CultureReset/cybercheck-orchestrator	—	CORE	The brain. Model routing, capability dispatching, you actually sell when you say you are selling.
CultureReset/cybercheck-marketplace	—	CORE	The app store. The index, the trust ladder, the spec.
CultureReset/cybercheck-web	—	USE	The public projection. Read-only, write-incompatible signed static half that the world sees.
CultureReset/App-build-	—	USE	App scaffolding. The generator for a new VAE, the surfaces. This is what a third-party developer uses.
CultureReset/API-layer-unified	—	USE	The seam. One contract in front of everything, nine different shapes. The place the capabilities are documented.
CultureReset/gcr-api-clean	—	USE	API, cleaned. Keep as the reference implementation above; fold what survives into API-layer-unified.
CultureReset/gcr-unified	—	STUDY	Earlier unification attempt. Read it for the details and forgot, then let it go. Two unified layers in one.
CultureReset/Admin-dashboard-main	—	USE	Your console. Fleet state, drift alerts, the real-time sit in front of on the Tuesday an app vendor's console.
CultureReset/Dashboards-users-	—	USE	The owner's console. Deliberately not the same answers "is my stuff right"; yours answers "what just broke".

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
CultureReset/Landin g-pages-	—	USE	The front door. Also the per-business public codebase, two jobs — worth keeping that way

01

The OS image and the update spine

The Red Hat trick, mechanically: the operating system is a signed OCI image, updates are image pulls, and a failed boot rolls itself back without a phone call. Everything in this layer exists so that shipping an update is a push and not a visit.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
✓ bootc-dev/bootc	Apache-2.0	CORE	The whole thesis in one binary. Boots a container system and updates it transactionally. Note that it's bootc now, not containers/bootc; the latter, which is exactly how you end up with a stale
ostreedev/ostree	LGPL-2.1	USE	The layer underneath bootc. Content-addressed atomic swap and rollback. You will rarely call it, but strange bootc behaviour is explained here.
osbuild/osbuild	Apache-2.0	USE	Image assembly. Turns a manifest into a bootable raw disk.
osbuild/bootc-image-builder	Apache-2.0	CORE	Container image → installable disk. The step between "I have something to put on a USB stick."
fedora-iot/greenboot	LGPL-2.1	CORE	Health check on boot, automatic rollback on failure. A component that makes remote updates to untrusted components a survivable business requirement in the face of catastrophe. Write real health checks, not ex
rauc/rauc	LGPL-2.1	STUDY	The A/B update alternative. Embedded-grade alternative to what you are doing. Correct choice if you ever ship some container runtime. Not both — pick one.
mendersoftware/mender	Apache-2.0	CAREFUL	OTA update platform with a managed server. The server is the business model; you are building the client-side delta logic, skip the rest.
canonical/ubuntu-core	—	STUDY	The snap-based immutable appliance. The closest thing to what you are doing. Study the store-and-forward model in particular — it is a working example of the correct way to do it at least of.
home-assistant/operating-system	Apache-2.0	STUDY	Read this one end to end before you commit. A system shipped to hundreds of thousands of non-technical users. A good update system, the recovery path and the "no downtime" discipline already solved in public.
home-assistant/core	Apache-2.0	STUDY	The integration model. Two thousand integrations with strangers without the project collapsing. The app store's future problem, already solved.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
ubblue-os/bluefin	Apache-2.0	STUDY	Universal Blue's daily driver. Proof that image usable by normal people today. Their build pipeline template.
ubblue-os/main	Apache-2.0	STUDY	The base images. How to layer your own things forking it.
ubblue-os/image-template	Apache-2.0	USE	Start here on day one. A working custom-build already wired. Saves a week of GitHub Actions.

The one decision in this layer

bootc or RAUC, and you cannot hedge. bootc gives you one artifact format from laptop to appliance and a registry you already run for everything else. RAUC gives you a smaller, older, more certain thing that works on hardware a container runtime would not fit on. Pick bootc for a mini-PC appliance; the rest of this catalog assumes you did.

02

The comparable, read closely

Omarchy is the project you were pointed at first, and it remains the closest thing to your opinionated-Linux-for-humans idea that has actually shipped and found an audience.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
basecamp/omarchy	MIT	STUDY	The original. DHH's opinionated Arch + Hyp the window manager; it is that taste, default any feature.
omacom-io/omarchy	MIT	STUDY	The current home. Check which org is cano anywhere public — this moved once already.
omacom-io/omarchy-pkgs	MIT	STUDY	Their package overlay. How a curated distro very small team.
omacom-io/omarchy-mirror	MIT	STUDY	Their mirror. Small, but it is the shape of dis will need — and cheaper to copy than to desi

Why it is study and not use

Omarchy is mutable Arch with excellent taste. You need immutable, signed and self-rolling-back, because your user is a restaurant owner and your update runs while the dinner rush is on. Take the taste and the positioning. Leave the architecture.

03

Boot chain and hardware root of trust

You said you want the least control possible. The way to hold almost none while still being able to prove the box is yours is to put the trust in hardware and in signatures, not in an account you can log into.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
torvalds/linux	GPL-2.0	USE	The kernel. Relevant decision: stay on in-tree need an out-of-tree module for the GPU, image simple. AMD's in-tree amdgpu is most of why silicon.
systemd/systemd	LGPL-2.1	USE	Init, services, sockets, sysex, credentials. Like the credentials system specifically — they so on a read-only OS without a mutable /etc.
tpm2-software/tpm2-tss	BSD-2-Clause	USE	TPM access. Seal the disk key to the measured data is unreadable if the drive walks out of the hold a key.
Foxboron/sbctl	MIT	USE	Secure Boot key management that a human boot a build step rather than a research project
OP-TEE/optee_os	BSD-2-Clause	STUDY	ARM TrustZone TEE. Only relevant if you go your needs.
fwupd/fwupd	LGPL-2.1	USE	Firmware updates on Linux, properly. The last Ignoring it means shipping known-vulnerable businesses.
fwupd/lvfs-website	GPL-2.0	STUDY	The firmware distribution service. A working you are building — signed artifacts, vendor a public metadata. Read the design documents cab file.

04

Apps on the box, and the naming trap

How software that is not the operating system gets onto and off of an immutable machine, and how the box describes what it has to a human being.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
flatpak/flatpak	LGPL-2.1	USE	Sandboxed desktop apps on an immutable C customer wants Chrome on it" that does not image.
flatpak/xdg-desktop-portal	LGPL-2.1	USE	The permission broker. How a sandboxed ap camera and the user answers. Your first-part door as third-party ones — that rule is worth claim about privacy.
PackageKit/PackageKit	GPL-2.0	USE	Distro-neutral package operations. The thin to so your UI never has to know what a pack
ximion/appstream	LGPL-2.1	CAREFUL	The freedesktop metadata standard for desc — icons, screenshots, summaries, categories the word. Between this and RHEL's AppStrea have two; a third called yours will cost you a conversations.
89luca89/distrobox	GPL-3.0	USE	Any distro's tooling inside a container on an own developers keep working on a machine a compiler.

05

The kiosk surface

The wall-mounted screen is where "a Linux computer made simple" either holds up or dies in front of everyone who works there. Nothing in this layer may ever show a terminal, an error dialog or a login prompt.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
cage-kiosk/cage	MIT	CORE	A Wayland compositor that runs exactly one desktop, no shortcut to a desktop, no way out product.
swaywm/sway	MIT	STUDY	The tiling compositor cage is built from. Re one thing you need, not before.
Quickshell/quickshell	LGPL-3.0	STUDY	QML shell toolkit. Relevant if the kiosk grow view into panels and overlays. Omarchy-adja
NoriginMedia/Norigin-Spatial-Navigation	Apache-2.0	USE	Arrow-key navigation for TV interfaces. You this is the unglamorous library that makes a Nobody notices it working; everyone notices

The rule for this layer

If a customer can reach a shell by accident, you have shipped a Linux box. If they cannot reach one at all, you have broken your own promise that they own it. The resolution is a deliberate, documented, slightly hidden way in — held down, typed in, never discovered by a dropped tray.

06

Containers, sandboxes and the isolation ladder

Third-party apps run on a machine sitting in a business's back office, holding that business's operational data. Isolation is not a feature here; it is the whole liability position.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
containers/podman	Apache-2.0	CORE	Daemonless, rootless containers with system writing. .container files that systemd manages beats Docker on an appliance. No daemon to
containers/buildah	Apache-2.0	USE	Build images without a daemon. Scriptable, podman.
containers/skopeo	Apache-2.0	USE	Move and inspect images between registries mirroring and promotion tool.
containers/image	Apache-2.0	STUDY	The library underneath skopeo and podman own promotion logic.
containers/storage	Apache-2.0	STUDY	The layer store. Same reason.
containers/bubblewrap	LGPL-2.0	USE	Unprivileged sandboxing primitive. What F1 cheapest real isolation you can put around a
google/gvisor	Apache-2.0	USE	A user-space kernel. Syscall interception, so gVisor instead of your kernel. The right default touches customer data.
google/nsjail	Apache-2.0	STUDY	Process jail. Lighter than gVisor, less complex tooling.
firecracker-microvm/firecracker	Apache-2.0	CAREFUL	MicroVMs with ~125ms boot. The strongest in hardware, and real operational weight. Justified from a developer you have never met — which eventually most of it.
google/crosvm	BSD-3-Clause	STUDY	The ChromeOS VMM. Read for how a consumer untrusted guests without the user ever hearing
microsandbox/microsandbox	Apache-2.0	STUDY	MicroVM sandboxing aimed specifically at a and smaller — watch it rather than depend on
bytecodealliance/wasmtime	Apache-2.0	USE	WebAssembly runtime with capability-based exactly the capabilities you hand it and cannot enforcement layer the Agent Plugins spec ex
extism/extism	BSD-3-Clause	CORE	The practical wrapper around Wasm plugin languages, plugins in any language that compile third-party code without third-party risk, the answer — and it is the difference between yo

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
e2b-dev/infra	Apache-2.0	STUDY	Open infrastructure for AI code sandboxes. answer to the same question. Read their isol
apple/container	Apache-2.0	STUDY	Apple's container tooling. Only interesting a platform you are positioned against is headi
apple/containerizat ion	Apache-2.0	STUDY	The framework under it. Same.

Where the ladder actually lands

First-party app: container + gVisor. Verified third-party app: the same, no exceptions and no shortcuts — identical treatment is the claim. Unverified community app: Wasm via Extism, with a capability list, or a microVM if it genuinely needs a filesystem. Never: a plugin that runs as a subprocess with the node's environment.

07

Registry, fleet and entitlement — the Red Hat machine

This is the part you are actually copying. Not the operating system: the apparatus that turns 'we publish updates' into a subscription business, with content on one side and entitlement on the other.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
project-zot/zot	Apache-2.0	CORE	A minimal OCI-native registry. Runs on the host, manages your OS images, your app bundles and your container images. Small enough to understand completely.
quay/quay	Apache-2.0	STUDY	Red Hat's registry. Overkill for you now. Realizes a security-account and vulnerability-feed model — that's what you're facing registry needs eventually.
pulp/pulp	GPL-2.0	STUDY	Content repository management: mirror, vendor, staging → production channel pattern. Used by people who ship to enterprises.
theforeman/foreman	GPL-3.0	STUDY	Lifecycle management for fleets of machines. Group and content-view concepts and implementation.
Katello/katello	GPL-2.0	STUDY	Foreman + Pulp + Candlepin together. This is how Red Hat actually does it. Satellite, and Satellite is how Red Hat actually does it. The single most useful thing in this stack.
candlepin/candlepin	GPL-2.0	CORE	Subscription and entitlement management. Manages what they're entitled to pull, when does it expire. Your business for updates; this is that business's logic.
flightctl/flightctl	Apache-2.0	CORE	Red Hat's edge device management for images. Manages device state, staged rollout, rollback. Built for a specific shape — one image, many unattended machines. The closest thing to a drop-in for your fleet plane.
cockpit-project/cockpit	LGPL-2.1	CAREFUL	Web admin for a Linux host. Useful for you, but you don't promise you do not want to make to a customer. If they break Cockpit, they can break the appliance, and then you're in trouble.
fleetdm/fleet	MIT	STUDY	Device management and reporting at scale. Fleet reporting model — you need fleet visibility with a good reporting model. This is the closest working example.
oras-project/oras	Apache-2.0	CORE	Push and pull arbitrary artifacts to an OCI registry. Images, packs, app bundles and model manifests all go to a mirrored, content-addressed place as your OCI mechanism for everything.
google/go-containerregistry	Apache-2.0	USE	The library for registry work in Go. crane is the CLI, but this will actually type.

The four pieces that matter

zot holds the bits. ORAS puts non-image things in it. Candlepin says who may pull. flightctl decides which devices get it and in what order. That is the subscription business, and every one of those four is open source and already written.

08

The pipeline and the policy gate

How a change becomes a signed artifact that four hundred machines are allowed to pull, with a gate that says no on its own without you in the room.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
tektoncd/pipeline	Apache-2.0	STUDY	Kubernetes-native CI. Correct if you are already not, and you should not start for this. Read the decomposition and do the same shape in which
konflux-ci/konflux-ci	Apache-2.0	STUDY	Red Hat's full supply-chain build service. Enforces — provenance, SBOM, policy — and system.
conforma/cli	Apache-2.0	CORE	The policy gate, formerly Enterprise Contract provenance and policy before an artifact is a is the machine that lets you say "nothing unsafe as a fact rather than a policy.
enterprise-contract/ec-cli	Apache-2.0	STUDY	The former name. Noted so an old link in a book different tool.
argoproj/argo-cd	Apache-2.0	CAREFUL	GitOps continuous delivery. Excellent, and I declared-state-in-git discipline; skip the clus

09

Supply chain: signing, transparency, SBOM, scanning

The evidence layer. Every claim you make about what is on the box has to be checkable by someone who does not trust you — including, eventually, a customer's insurer.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
sigstore/cosign	Apache-2.0	CORE	Sign and verify container images and arbiters via OIDC means you are not managing a private key where signing schemes usually die. Non-negative attestation reaches a customer machine.
sigstore/rekor	Apache-2.0	CORE	The transparency log. An append-only public log of when. Put constitution versions and append-only not just OS images — those are the claims that are verified. It is also your defence on the data that changed a business's data.
sigstore/sigstore	Apache-2.0	STUDY	The client libraries. For when signing has to be done in code.
theupdateframework/python-tuf	Apache-2.0 / MIT	STUDY	TUF: the formal answer to compromised updates. Separation, key rotation, rollback and freeze covers most of it in practice; read TUF for the details asked about by the first serious customer.
anchore/syft	Apache-2.0	USE	Generate an SBOM from an image or directory. Every build from the first build, because retroactive SBOM of releases is a month of work.
anchore/grype	Apache-2.0	USE	Scan that SBOM for known vulnerabilities. If you have your "we patch it" claim made mechanical.
aquasecurity/trivy	Apache-2.0	USE	Broader scanner — images, filesystems, IaC, etc. One in CI and the other as an independent check to tell you little.
trufflesecurity/trufflehog	AGPL-3.0	USE	Finds committed secrets, and verifies whether they are safe. Every third-party app bundle before it enters your CI. As fine as a CI tool you execute, a problem if you distribute.
aboutcode-org/scancode-toolkit	Apache-2.0	USE	License and origin detection across a codebase. Hundreds and eighty projects into hardware by the GPL component someone vendored into your code, before a customer's lawyer does.
CycloneDX/cyclonedx-cli	Apache-2.0	USE	SBOM format conversion and merging. You get 40 per released image, not forty.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
spdx/tools-python	Apache-2.0	STUDY	The other SBOM standard. Emit whichever tooling to convert exists, so do not hold a rel

The claim this layer buys you

"I cannot see your data" is architecture. "You can verify what I sent you" is signing plus transparency. The second is the one that survives a hostile question, because the person asking can check it without your cooperation.

10

The execution layer — remote control of a real phone

The core of the product and the only row in the whole catalog with no upstream that does the whole job. Everything here is a part of the mechanism: see the screen, read the tree, act, and prove the act landed.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
✓ Genymobile/scrcpy	Apache-2.0	CORE	Mirror and control an Android device over USB. Low latency, rock solid, enormous what makes "they can see their own phone on work instead of a product.
NetristTV/ws-scrpcy	MIT	USE	scrcpy in a browser over WebSocket. The only shipping a native app to anyone.
✓ openatx/uiautomator2	MIT	CORE	Python control of Android's UIAutomator: find elements, tap, type, wait. This is the exact what ghost/device.py was reimplementing library and delete about three hundred lines
openatx/adbutils	MIT	USE	ADB from Python without shelling out. The use it directly for device discovery, install, se
✓ mobile-dev-inc/maestro	Apache-2.0	CORE	Declarative YAML flows for mobile UI, with implicit waits that actually work. The hardest in this whole build is "wait until the screen is has solved it in public. Verified today: still App releasing every few weeks.
mobile-dev-inc/maestro-studio	Apache-2.0	CORE	Point at an element, get a selector, record a flow — the thing you use on the Tuesday an app v you have to re-map before the fleet notices. I ghost/recorder.py was rebuilding.
appium/appium	Apache-2.0	STUDY	The old guard of mobile automation. Heavier driver architecture is the reference for support later.
appium/appium-inspector	Apache-2.0	USE	A GUI for inspecting a live app's element tree a selector matches nothing and the screensh
appium/appium-uiautomator2-driver	Apache-2.0	STUDY	Their Android driver. Read it when uiautom cannot explain.
google-research/android_world	Apache-2.0	STUDY	A benchmark of 116 tasks across 20 real Android regression suite. Also use its published success on what unattended UI automation achieves fingerprint gate exists to work around.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
OpenBMB/AgentCPM-GUI	Apache-2.0	STUDY	An on-device GUI agent model. Interesting to survive a layout change without a human re-mechanism; your v1 advantage is that you do button.
bytedance/UI-TARS	Apache-2.0	STUDY	A GUI agent model with real benchmark num further along. Read the failure analysis rather
microsoft/OmniParser	MIT	STUDY	Turns a screenshot into structured, labelled when there is no accessibility tree — for example game-engine UI.
similar-ai/Agent-S	Apache-2.0	STUDY	An agent framework for computer use. Read design.
mobile-next/mobile-mcp	Apache-2.0	USE	Mobile control exposed as an MCP server. Not a projection. This is how an assistant asks you something without being handed ADB.
✓ droidrun/mobilerun	Apache-2.0	STUDY	LLM-agnostic mobile agent: inspect UI state swipe, type, plan. Verified today as a live proc to your executor — read it for their tool surface their complexity comes from letting a model
✓ callstackincubator/agent-device	MIT	USE	Mobile automation and verification for code and typed Node API across iOS, Android, TV Note the correct org: it is callstackincubator word <i>verification</i> in their own description is to how they do it.
sktyou/OpenGUI	—	STUDY	An Android GUI agent framework for phone small and young. Read, do not depend.
yashab-cyber/opendroid	—	CAREFUL	Appeared in the session's search results. I checked today — GitHub is unreachable from this environment not a part, until you have opened it yourself.
waydroid/waydroid	GPL-3.0	CAREFUL	Android in a container on Linux. Tempting: defeats the entire premise — the customer's cloud logged-in accounts, a real device they can pick up, wrong for the product.
block/trailblaze	Apache-2.0	STUDY	Android UI automation aimed at agents. Small path.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
google/device-infra	Apache-2.0	STUDY	Google's own device lab infrastructure. Reads on a rack and the problem becomes logistics
textbee/textbee	GPL-3.0	USE	Turns an Android phone into an SMS gateway, the text-message marketing channel, sent from a number, with no Twilio in the path and no phone number. GPL-3.0 before it goes in a shipped image.
google/artemis	—	SKIP	Appeared in the session as a mobile-automation tool. verify it today and cannot confirm the repository. as. Do not put it in a plan until you have looked

The row with no upstream

Fingerprint-gated, independently verified remote control is the one job in this catalog that nothing here does. Every project above will happily tap a button and report success. None of them refuse to act because the screen is not the screen they were mapped against, and none read the result back on a different path to prove it landed. That gate and that read-back are the product. Everything else is a part you did not have to build.

11

The other two execution paths

The verifier must not be the executor. If you write through Android and read back through Android, you have built self-certification with extra steps. These are the independent paths.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
microsoft/playwright	Apache-2.0	CORE	Browser automation that is reliable enough vendors that have a usable web interface, an independent read-back path that proves an A landed.
microsoft/playwright-mcp	Apache-2.0	USE	Playwright as an MCP server. Same projectio assistant asks, it does not drive.
browser-use/browser-use	MIT	CAREFUL	Let a model drive a browser. Genuinely usef quickly. Keep it in the factory, out of the run on a customer's live booking system is the ex design avoids.
trycua/cua	MIT	STUDY	Computer-use agents in sandboxed VMs. Re
OpenHands/OpenHands	MIT	STUDY	Formerly OpenDevin. Read the execution-ar the coding focus.
openinterpreter/open-interpreter	AGPL-3.0	CAREFUL	Natural language to executed code, locally. I makes it a distribution problem for a box you
ydotool/ydotool	AGPL-3.0	CAREFUL	Input automation on Wayland. Sometimes tl stubborn desktop app. Check the licence aga

Say it in the manifest

Each capability should name its write path and its verify path, and the two must differ. "Wrote via Android, verified via browser" is a sentence a customer understands and a claim a sceptic can test. It is also the difference between a run record and a receipt.

12

Their screen, on their phone

The feature that makes an owner feel they own the thing: pick up an iPhone, see the Android device sitting in the back office, take over.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
LizardByte/Sunshine	GPL-3.0	CAREFUL	Self-hosted low-latency desktop streaming k ships on hardware you sell — read it properly
moonlight- stream/moonlight-qt	GPL-3.0	STUDY	The client side. Same licence note.
Genymobile/scrcpy	Apache-2.0	CORE	Listed again deliberately. For the phone spec scrcpy is a simpler and better answer than a Use the streaming pair only if you ever need screen.

13

Agent protocols and the orchestration layer

You are selling the harness. This is the wiring the harness is made of — and the standards question is now settled enough to build on.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
modelcontextprotocol/modelcontextprotocol	MIT	CORE	MCP, the specification. The way a customer's to an assistant without leaving their machine on their computer' promise is this protocol, i
modelcontextprotocol/servers	MIT	USE	Reference servers. Read three, then write yo useful than the code.
modelcontextprotocol/python-sdk	MIT	USE	Python SDK. The node speaks this.
modelcontextprotocol/typescript-sdk	MIT	USE	TypeScript SDK. The surfaces speak this.
✓ a2aproject/A2A	Apache-2.0	CORE	Agent-to-agent communication. This is the v developer story: the developer's agent talks t answer crosses, the data does not. Verified to has converged under Linux Foundation gove folded into this effort rather than competing
agentclientprotocol/agent-client-protocol	Apache-2.0	STUDY	The editor-to-agent protocol. Adjacent; read
IBM/mcp-context-forge	Apache-2.0	CORE	ContextForge: an MCP gateway and registry. servers behind one endpoint with auth, rate You will have dozens of MCP servers per nod becomes unmanageable around server num
i-am-bee/beeai-framework	Apache-2.0	CAREFUL	IBM's multi-agent framework, now under th today: real governance, real backing. Still m agent/workflow decomposition, and keep yo independent of it so a breaking release is an
i-am-bee/beeai-platform	Apache-2.0	STUDY	The platform around the framework. Discov for agents from different frameworks. Closes orchestrator's job.
i-am-bee/beeai-code-interpreter	Apache-2.0	STUDY	Sandboxed code execution for agents. Comp choosing.
NousResearch/hermes-agent	Apache-2.0	STUDY	An agent harness built around tool use. Rea

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
langchain-ai/langgraph	MIT	CAREFUL	Graph-structured agent workflows with che machine is the good idea. Take it; resist the r is a dependency surface you do not need on
block/goose	Apache-2.0	STUDY	A local, extensible agent. Read the extension your VAPP contract has to do.

Where the dumb agent lives

MCP is how it reaches data. A2A is how it talks to someone else's agent without handing over the data. ContextForge is how you survive having thirty MCP servers per node. Everything else in this section is reading.

14

Workflow, events and durable execution

A flow that taps four screens across two apps and dies halfway has to be explainable, resumable and — where it cannot resume — compensable. That is a durability problem, not an automation one.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
temporalio/temporal	MIT	CORE	Durable execution: workflows survive crash with full history. The compensating-action pattern applied to two of four destinations is worse than what is exactly what this is built for. The run history is free.
restatedev/restate	BUSL-1.1	CAREFUL	Durable execution, lighter to operate. Note that it is available, not open source, with a delayed commercial model decision you would be adopting, not just a
nats-io/nats-server	Apache-2.0	USE	Messaging with JetStream persistence, in a small package. Right size for a box in a restaurant. Kafka is the elephant of magnitude.
apache/kafka	Apache-2.0	SKIP	Not for an appliance. Listed because it came up in the why you said no is worth a line.
apache/flink	Apache-2.0	SKIP	Same. Stream processing at a scale you will not want to manage.
ansible/ansible	GPL-3.0	USE	Agentless configuration and orchestration. The factory bench, not something that runs on a
ansible/awx	Apache-2.0	STUDY	The upstream of Automation Controller. Real credential model.
ansible/event-driven-ansible	Apache-2.0	STUDY	Rules that fire actions from events. The shape of a re-map-queued pipeline.
activepieces/activepieces	MIT	STUDY	Open-source automation with an MCP story. The model — it is close to your VAPP shape and gets
ComposioHQ/composio	Apache-2.0	STUDY	A large managed catalogue of tool integrations. They handle auth per-tool; that problem will
node-red/node-red	Apache-2.0	STUDY	Flow-based programming with a visual editor. About what non-programmers can and cannot do. One day an owner wants to change a flow.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
n8n-io/n8n	Sustainable Use	SKIP	Not open source. The Sustainable Use Licence is not open source. It gets called open source constantly as a marketing advice. Using it inside a product you sell is a bad idea. You do not want to have late.

15

Models and inference — the terabyte of specialists

A terabyte of specialised models on the box, one endpoint in front of them, and something that swaps them fast enough that a customer never feels it.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
ggml-org/llama.cpp	MIT	CORE	The engine room. Quantised inference on or OpenAI-compatible server built in. MIT, no s unified-memory part you picked.
mostlygeek/llama-swap	MIT	CORE	Hot-swaps models behind one endpoint, each the specific piece that makes 'a terabyte of sp being a directory of files. Process isolation m cannot take the endpoint down.
ggml-org/whisper.cpp	MIT	USE	Speech recognition, same engine family. Use runtime rather than two.
vllm-project/vllm	Apache-2.0	CAREFUL	High-throughput serving. Built for many co GPUs. You have one business per box. Right a wrong answer for the appliance.
ollama/ollama	MIT	CAREFUL	The easiest local-model experience there is llama.cpp. Perfect for your laptop this week. one layer down so you control quantisation, behaviour yourself.
BerriAI/litellm	MIT	USE	One API shape in front of every provider, lo claim, in code: the customer can point at you OpenAI key, and nothing upstream changes.
open-webui/open-webui	BSD-3-Clause	STUDY	A polished local-model UI. Read the model-r a UI problem you would otherwise solve bad
huggingface/huggingface_hub	Apache-2.0	USE	Fetch and cache models programmatically. download path — and worth mirroring into y customer's box does not depend on someone
instructlab/instruclab	Apache-2.0	STUDY	Community model tuning with synthetic da 'the restaurant vertical needs its own adapte without a research team.
ibm-granite/granite-guardian	Apache-2.0	USE	A small model that checks other models' inp Cheap insurance on a device that writes to a
ibm-granite/granite-speech-models	Apache-2.0	STUDY	Speech models, Apache-2.0. An alternative t licence.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
ibm-granite/granite-vision-models	Apache-2.0	STUDY	Vision models. Relevant for reading a screen tree fails.
ibm-granite/granite-tsfm	Apache-2.0	STUDY	Time-series foundation models. Demand for
meta-llama/llama-stack	MIT	STUDY	A standardised API surface for inference, sa Responses API shape whether or not you add the common denominator.
kserve/kserve	Apache-2.0	SKIP	Kubernetes model serving. Wrong deployme
llm-d/llm-d	Apache-2.0	SKIP	Distributed serving on Kubernetes. Same.
ROCm/ROCm	MIT	USE	AMD's compute stack. The reason the hardw choice are the same decision. Verify your ex exact ROCm version before you buy a pallet.
NVIDIA/Personal-AI-Router	Apache-2.0	STUDY	Routing between local and remote models. I need the same decision made on cost, privac
agentscope-ai/QwenPaw	Apache-2.0	STUDY	An agent runtime worth a read, not a dependen
xai-org/grok-1	Apache-2.0	STUDY	The weights release. Relevant only as eviden the company you are positioning against.

The swap trick worth knowing

Loading a different model costs five to eight seconds. Swapping a LoRA adapter on a loaded base costs roughly two hundred milliseconds. If your verticals can be adapters over one base rather than twenty separate models, the box feels instant and the terabyte becomes a library rather than a queue.

Voice

Wake word, endpointing, transcription, speech. Four small models and one protocol that lets them be swapped independently. This is also the demo people remember.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
rhasspy/wyoming	MIT	CORE	The protocol that lets each voice piece be a s TTS without touching the wake word. Home production across a very large install base.
dscripka/openWakeWo rd	Apache-2.0	USE	Wake word detection, trainable on your own training run, not a licensing negotiation.
snakers4/silero-vad	MIT	USE	Voice activity detection. Tiny, fast, and the d that knows when you stopped talking and or
SYSTRAN/faster- whisper	MIT	USE	Whisper via CTranslate2 — several times fas practical transcription choice.
✓ rhasspy/piper	MIT	SKIP	Archived 6 October 2025. Verified today. It is everywhere and still in a great many build g
✓ OHF-Voice/piper1- gpl	GPL-3.0	CAREFUL	Where Piper's development actually moved the licence change from MIT to GPL-3.0. If yo because it was MIT, that reason no longer ex kind of thing that is expensive to discover af hardware.
hexgrad/kokoro	Apache-2.0	CORE	82M parameters, genuinely good voice, Apa now: small enough for the box, clean enough
speaches- ai/speaches	MIT	USE	An OpenAI-compatible server for speech-to- familiar API in front of the whole voice stack
pipecat-ai/pipecat	BSD-2-Clause	USE	Real-time voice conversation orchestration barge-in. The things that separate a voice de does not infuriate callers.
livekit/livekit	Apache-2.0	USE	WebRTC infrastructure, self-hostable. The t agent has to answer an actual phone call.
livekit/agents	Apache-2.0	STUDY	Their agent framework on top. Read the tur
fonoster/fonoster	MIT	STUDY	Open-source programmable telephony. The question you already asked out loud — why a

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
home-assistant/hassil	Apache-2.0	USE	Template-based intent matching. Most voice are eight fixed sentences. Matching them with cheaper and more predictable than asking a hallucinates a reservation.

17

The data plane

Private truth on the node, signed projections on the read-only edge, and a backup the customer controls and you cannot read.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
postgres/postgres	PostgreSQL	CORE	The node's database. One node per business, anywhere — the isolation is the deployment, not a WHERE clause.
pgvector/pgvector	PostgreSQL	USE	Vectors inside Postgres. One database to back up, one reason about.
asg017/sqlite-vec	Apache-2.0 / MIT	USE	Vector search in SQLite. For the small edge cases, no much machinery.
benbjohnson/litestream	Apache-2.0	USE	Continuous streaming replication of SQLite recovery for the small database, with no server.
restic/restic	BSD-2-Clause	CORE	Encrypted, deduplicated backup to a destination. Their NAS, their bucket, their key. They are your, you still hold nothing. This one component is the 'liability' position made real.
meilisearch/meilisearch	MIT	USE	Fast, typo-tolerant search with almost no options. size for 'who has crab legs nearby' across one.
qdrant/qdrant	Apache-2.0	STUDY	A dedicated vector database. Only if pgvector is not it will not for a long time.
opensearch-project/OpenSearch	Apache-2.0	SKIP	Too heavy for a box. Noted so the question is not.
supabase/supabase	Apache-2.0	CAREFUL	Postgres plus auth, storage and realtime, selling a large surface. If you adopt it, adopt it deliberately rather than letting it arrive one feature at a time.

The thing you cannot add later

Provenance. If you write canonical values for six months and then decide you want to know where each one came from, you have six months of data with no origin and no way to reconstruct it. The observation table goes in before the first ingestion runs, not after the first dispute.

18

Getting the data in — email as the universal API

Every vendor already emails the business: the booking confirmation, the daily sales summary, the payout notice. That inbox is an integration nobody can revoke.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
apache/tika	Apache-2.0	USE	Extracts text and metadata from a thousand unglamorous front door for every PDF a ven
docling-project/docling	MIT	CORE	Turns PDFs into structured documents — tal mangled text. A daily sales summary is a tab component that makes it a row in your data
google/langextract	Apache-2.0	USE	Structured extraction from unstructured tex recorded. That last part is the point: every c the exact sentence it came from.
google/magika	Apache-2.0	USE	Reliable file-type detection. The first check outside.
google/schema-dts	Apache-2.0	USE	Typed schema.org for TypeScript. How the machine-readable, which is how a business assistant's answer instead of an index's.

Rank the setup paths by friction

First: add your address as a second notification email inside the vendor's own settings — Toast, Square and FareHarbor all allow it, and Gmail is never involved. Second: a Gmail filter and a verified forwarding address, two minutes in a UI they already know. Last resort: an Apps Script. "Run this script" kills conversion at the exact moment the product needs to feel like nothing.

19

Identity, authorization, policy, secrets

One login, many apps — and a permission model the Agent Plugins specification openly says it does not have. This layer is where your store stops being a copy of someone else's.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
zitadel/zitadel	Apache-2.0	CORE	Identity and access management, self-hosted 2.0. Lighter than Keycloak to run and operate apps' with nothing phoning home.
keycloak/keycloak	Apache-2.0	CAREFUL	The heavyweight, Red Hat's SSO. More capable only if you need something Zitadel lacks, and
authzed/spicedb	Apache-2.0	CORE	Google Zanzibar-style fine-grained authorization permissions — this user, this business, this context enforcement point for capability lists, and the quietly read something it was not granted.
openfga/openfga	Apache-2.0	STUDY	The CNCF alternative. Same model, different never run both.
open-policy-agent/opa	Apache-2.0	USE	Policy as code, evaluated anywhere. The gate capability call is allowed given the constitution time of day.
openbao/openbao	MPL-2.0	CORE	The open fork of Vault. Secrets, dynamic credentials, customer's vendor logins live here, on their machine — not in a config file and not in your cloud.

The gap you are filling, stated plainly

The Agent Plugins specification says, in its own words, that it contains no explicit capability or permission system. Grok's install command is `--trust`, and trust is binary. Fine on a developer's laptop; not fine on a machine holding a business's operational data. SpiceDB plus OPA plus Extism is the answer, and it is the most defensible thing you are building.

20

Reaching the box without owning it

You need to support a machine in someone's back office without a standing door into their network — and without a VPN account you could be compelled to use.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
juanfont/headscale	BSD-3-Clause	USE	A self-hosted control plane for Tailscale client coordination, the customer runs the node, and the control plane is end encrypted between devices.
tailscale/tailscale	BSD-3-Clause	STUDY	The client, and the hosted service. Read the docs to host the control plane.
netbirdio/netbird	BSD-3-Clause	STUDY	The all-in-one alternative. Simpler to stand up and operate.

Support access is a design decision, not a default

The strongest version: no standing access at all. The customer grants a session, it expires, and the grant is recorded in the transparency log. That is harder to build and much easier to say out loud in a sales conversation — and it is the same argument you are already making about data.

21

Money

Zero percent of the platform, one percent of the payment volume, and a spread you renegotiate as the volume grows. That arrangement needs metering you can defend.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
getlago/lago	AGPL-3.0	USE	Open-source usage-based billing and metering. here because you run it as a service rather than make that call deliberately.
killbill/killbill	Apache-2.0	STUDY	Mature subscription billing. Heavier, Apache cases already found by other people.
dubinc/dub	AGPL-3.0	STUDY	Link attribution. The referral and co-op loyalty attribution that survives a click; this is a work in progress.
candlepin/candlepin	GPL-2.0	CORE	Listed again on purpose. Lago meters usage, entitlement — whether this box may pull this together, and conflating them is how subscription to answer 'is this customer paid up' from a single

The number to watch

With destination charges the platform carries dispute liability by default, plus negative balances when a business closes holding deposits. Budget 0.1–0.3% of volume for disputes and reserves. That is what turns a one-point spread into roughly seven tenths of one.

22

The app store layer

The package format is settled, the distribution model is proven, and the trust ladder already exists in three different forms. None of this needs inventing — which is Part VI's whole argument.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
✓ agentplugins/agent-plugins-spec	—	CORE	Agent Plugins 1.0.0 — the package format. V-backed than the session thought: the technique Amazon, Cursor, Microsoft, OpenAI and Vercel Foundation. plugin.json, a strict field list, containment, PLUGIN_ROOT/PLUGIN_DATA, manifest rides in extensions.io.anextger
agentskills/agentskills	—	CORE	Agent Skills — the unit inside the package. S and progressive disclosure: about a hundred description loaded for every skill at startup, resources only on demand. That first tier is y description field is the trigger rule.
neondatabase/agent-skills	Apache-2.0	STUDY	A real, well-built skill set. Read the description written as phrase lists that route, which is th
✓ vercel/skills	—	USE	The installer pattern, verified today. Vercel's forty-plus agents using a provider registry, a .skill-lock.json that compares directory That lock-file-and-tree-SHA design is what yo comes from a company on the specification's
junior/skilla	—	CAREFUL	The installer the session identified, with its not re-verify it today — GitHub is unreachable surface in search. Check it yourself; if it has above is the better-supported pattern anywa
xai-org/plugin-marketplace	—	STUDY	Index, not store. The catalogue entry points at a pinned full-length commit SHA. xAI nev model costs almost nothing to run and puts l read it closely, it is the cheapest distribution
xai-org/grok-build	—	STUDY	The build tooling around it. Read the manife
superagent-ai/grok-cli	MIT	STUDY	A third-party client. Evidence of what the ec is published.
vercel/vercel-plugin	—	STUDY	A real plugin in the wild. Read one before yo
getsentry/sentry-for-ai	—	STUDY	Another. Two examples is enough to see the

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
redhat-openshift-ecosystem/redhat-marketplace-operators	Apache-2.0	STUDY	How certified content is submitted, reviewed submission pipeline for a trust ladder, already who run one.

The trust ladder, borrowed from people who run one

Red Hat operates four distinct stores, and Automation Hub has three tiers: community content, certified content, and a private hub the customer controls. Map yours to community → verified → the business's own private index. The pinned full SHA is what makes any of it enforceable, because a tag can be moved and a forty-character commit hash cannot.

23

Surfaces — what the customer and the world actually see

Four surfaces, one record behind them. The owner's console, the kiosk, the public page, and the agent endpoint.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
puckeditor/puck	MIT	USE	A visual editor for React, embeddable in your app. It rearranges their public page without a developer building a page builder from scratch.
shadcn/ui	MIT	USE	Components you copy into your codebase rather than upstream that can break you, no version to control. Product with a ten-year support horizon.
flutter/flutter	BSD-3-Clause	CAREFUL	One codebase for the owner's phone app. React matters; a second language and toolchain for web app first and be honest about whether a phone app.
microsoft/PowerToys	MIT	STUDY	Listed because it came up. Read PowerToys for the box's command surface, then move on.

24

Read, do not adopt — the IBM and Red Hat shelf

You said Red Hat is the model and Watson was open source. Both true. These are worth reading for how a platform business is structured, and worth not adopting, because each one assumes an operating environment you do not have.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
IBM/watsonx-developer-hub	Apache-2.0	STUDY	How a large vendor presents an agent platform onboarding path — it is the thing you are confronted with in a developer's first hour.
IBM/watsonx-data	—	STUDY	The data layer. Read the governance vocabulary words.
opendatahub-io/opendatahub-operator	Apache-2.0	STUDY	The upstream of Red Hat's AI platform. Kubernetes component decomposition, which is a good example of what is made of.
cockpit-project/cockpit	LGPL-2.1	STUDY	Listed twice deliberately. As a study item it is a UI that manages a Linux host without preten-

What the IBM and Red Hat reading is actually for

Not the code. The structure: content in one system, entitlement in another, certification as its own track, metering separate from billing, and a policy gate that can say no without a human. That structure is the business you described, and it is documented in public by a company that makes billions running it.

25

The live traps

Things that will cost you a week or a lawyer if you find them late. Every one of these is currently a top search result for something you will search for.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
✓ rhasspy/piper	MIT, archived	SKIP	Archived October 2025 , and development m... Verified today. Every tutorial still points her...
n8n-io/n8n	Sustainable Us	SKIP	Not open source , despite being described th... Commercial hosting is restricted.
ximion/appstream	LGPL-2.1	CAREFUL	The name collision. This project, RHEL's App... anything of yours called AppStream makes t... while renaming is a find-and-replace.
ollama/ollama	MIT	CAREFUL	Not a trap, a boundary. Ship llama.cpp; deve... wrapper means debugging someone else's de...
restatedev/restate	BUSL-1.1	CAREFUL	Source-available, not open source. BUSL cor... delay; know the date before you depend on i...
trufflesecurity/tru fflehog	AGPL-3.0	CAREFUL	AGPL. Fine as a tool your CI runs. A distribut... into something you ship.
openinterpreter/ope n-interpreter	AGPL-3.0	CAREFUL	AGPL. Same reasoning.
getlago/lago	AGPL-3.0	CAREFUL	AGPL. Run it as a service and it is fine; bundl...
LizardByte/Sunshine	GPL-3.0	CAREFUL	GPL-3.0 on hardware you sell brings the anti... play. Read them properly before this become...
textbee/textbee	GPL-3.0	CAREFUL	GPL-3.0. Same question, and this one sits in t... you already want.
89luca89/distrobox	GPL-3.0	CAREFUL	GPL-3.0. A developer tool rather than shippe... — but note it.
waydroid/waydroid	GPL-3.0	CAREFUL	GPL-3.0, and wrong for the product anyway. the point.
google/artemis	—	SKIP	Unverifiable from here. It was in the session... today. Anything you cannot open is not a par...
google/CAGE	—	SKIP	Same. Confirm before it appears in a plan.
yashab- cyber/opendroid	—	CAREFUL	Same. A lead until you have read it.

PROJECT	LICENCE	VERDICT	WHAT IT IS, AND THE JOB IT DOES HERE
junior/skilla	—	CAREFUL	Same. The Vercel skills CLI is the better-suppr

Why this section exists

Licences change, projects get archived, and orgs move. Three of the entries above changed after they were first written down in this project — Piper archived and relicensed, bootc moved orgs, agent-device is under callstackincubator rather than callstack. Re-check anything load-bearing before a build, not after a customer calls.

The order

Nine steps, and the order matters more than the list

Anyone can clone two hundred and twenty-six repositories in a weekend. The useful question is which three are working by Friday.

- 01

scrcpy — get a real Android phone mirrored on a Linux box with working input. One afternoon, and the first time the idea stops being a description.
- 02

uiautomator2 — dump the view tree of the Google Business Profile app and tap a button from Python. Then delete ghost/device.py.
- 03

Maestro and Maestro Studio — record one flow by pointing at elements. Compare what their waiting logic does to what yours was going to do.
- 04

The test that decides everything — install the real apps a Destin charter or a restaurant actually uses. Turn on an accessibility service. See what still runs. This is the one assumption underneath the entire business, and it has still not been run.
- 05

llama.cpp and llama-swap — one endpoint, two models, hot swap on demand. An hour, and the orchestrator has an engine room.

- 06 **wyoming + openWakeWord + silero-vad + faster-whisper + kokoro** — wire five small things together and say a sentence to your box. This is the demo people repeat to other people.

- 07 **Playwright** — write the same fact through Android and read it back through the browser. The moment those two agree on a different path, you have the product.

- 08 **home-assistant/operating-system** — read the update and recovery design end to end before committing to your own.

- 09 **bootc + greenboot** — build one image, boot it, break it on purpose, watch it roll itself back. Now you have a company rather than a script.

Steps one to four are the product

Five and six are the demo. Seven is the defensible part. Eight and nine are the company. Step four is the only one that can end the project, which is an argument for running it this week rather than after another document.

Compiled from the full working record. Two hundred and twenty-six projects, twenty-six layers, two hundred and forty-four entries — some listed twice where a project is both a part and a trap. Exactly one job in this catalogue has no upstream: fingerprint-gated remote control with independent read-back. Everything else is a part you did not have to build.